NOUR EMAD ABDALLAH

Data Science & Artificial Intelligence Specialist

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PROFESSIONAL SUMMARY

Data Science & AI undergraduate with hands-on experience in predictive modeling, computer vision, and end-to-end machine learning pipeline development. Skilled in building production-ready AI systems that combine rigorous statistical analysis with interactive applications. Strong foundation in Python, SQL, feature engineering, model validation, and real-time data processing.

EDUCATION

Bachelor of Science in Data Science & Artificial Intelligence

Alexandria National University, Alexandria, Egypt | Expected June 2027

Relevant Coursework: Machine Learning, Expert Systems & Knowledge Representation, Data Structures & Algorithms, Operating Systems, SVM, Regression Analysis, DBMS, Numerical Analysis

TECHNICAL SKILLS

Programming Languages & Core Technologies: Python (Advanced), SQL (SQLite, PostgreSQL), MATLAB, JavaScript, HTML5, CSS3

Machine Learning & AI: YOLOv8, Object Detection & Tracking (ByteTrack), SVM / SVR, Decision Trees, Random Forests, Particle Swarm Optimization (PSO), PCA, Regression Analysis, Neural Networks, Unsupervised Learning, Recommender Systems

Libraries & Frameworks: Scikit-Learn, Pandas, NumPy, SciPy, OpenCV, Streamlit, PyQt5, Tkinter, Matplotlib

Tools & Environments: Git, GitHub, Jupyter Notebook, VS Code, AWS (EC2, S3, RDS, VPC, Cloud Foundations)

PROJECTS

Intelligent Traffic Surveillance & Object Analysis System

Python · YOLOv8 · OpenCV · ByteTrack · NumPy

- Engineered a real-time computer vision pipeline using YOLOv8 for high-accuracy vehicle detection and multi-object tracking across crowded traffic scenes.
- Integrated ByteTrack to maintain continuous object identities under occlusion, camera jitter, and high-density environments.
- Designed spatial analytics module covering speed estimation, directional flow counting, anomaly detection, and structured diagnostic logging.

Medical Image Registration using Particle Swarm Optimization (PSO)

Python · PSO · OpenCV · SciPy · PyQt5

- Developed a medical image alignment engine using Particle Swarm Optimization for CT/MRI scan registration.
- Built a multi-metric evaluation suite (MSE, SSIM, NCC, Mutual Information) to benchmark alignment quality across modalities.
- Created a desktop GUI enabling parameter tuning, convergence visualization, and automated PDF report generation.

Expert Heart Disease Risk Predictor

Python · Streamlit · Scikit-Learn · Pandas · NumPy

- Built a hybrid clinical decision-support system that pairs ML predictions with an interpretable expert-reasoning layer.
- Trained and optimized a Decision Tree Classifier achieving ~80% accuracy on real clinical datasets.
- Deployed an interactive Streamlit portal supporting patient data input, diagnostic validation, and exportable PDF medical summaries.

King County House Price SVR Predictor

Python · SVR · Scikit-Learn · Streamlit · Pandas

- Engineered an end-to-end SVR regression pipeline for real-estate price prediction with robust preprocessing.
- Applied outlier handling, spatial feature extraction, log-transformations, and PCA to improve model generalization.
- Deployed a Streamlit application supporting dataset upload, hyperparameter tuning, and interactive residual visualization.

Operating Systems Algorithms GUI Simulator

Python · PyQt5 · Tkinter · Matplotlib · OOP

- Built a desktop educational tool simulating CPU scheduling (FCFS, SJF, Round Robin) and memory/page-replacement algorithms (FIFO, LRU, Clock).
- Rendered interactive Gantt charts and performance metrics for side-by-side algorithm comparison.
- Applied object-oriented design to decouple algorithm logic cleanly from the GUI visualization layer.

Healthcare Risk Analysis — Canonical Correlation Analysis (CCA)

Python · MATLAB · Pandas · SciPy · Matplotlib

- Applied CCA to quantify relationships between lifestyle variables and clinical wellness indicators.
- Implemented parallel analytical pipelines in Python and MATLAB to cross-validate canonical weights and loading matrices.
- Conducted significance testing via Wilks' Lambda and permutation analysis to assess statistical robustness.

CERTIFICATIONS

AWS Academy Graduate — Cloud Foundations Training Badge

Amazon Web Services (AWS Academy) | Issued December 2025

AWS fundamentals, EC2, S3, VPC, RDS, cloud security, architecture, and pricing.

Google Advanced Data Analytics Professional Certificate

Coursera | In Progress

Python, R, SQL, Statistical Analysis, Regression Models, Machine Learning, Tableau.

Machine Learning Specialization

DeepLearning.AI & Stanford University | In Progress

Supervised ML, Advanced Learning Algorithms, Neural Networks, Unsupervised Learning, Recommender Systems.

CORE COMPETENCIES & LANGUAGES

Professional Skills: Problem Solving | Analytical Thinking | Attention to Detail | Critical Thinking | Research | Self-Learning | Time Management | Technical Communication | Collaborative Development | Presentation Skills

Languages: Arabic — Native | English — Fluent / Professional Working Proficiency